# Quotation: TEST INPUT - readiness trace pump panel

Job number: TEST-001

Boards: PP1

Currency: JOD

| Description | Qty | Unit | Unit price | Total |
|---|---|---|---|---|
| Circuit-breaker 4P C100<br>-Q1 | 1 | pcs | not priced | - |
| Circuit-breaker 3P C16<br>-Q2 | 1 | pcs | not priced | - |
| Residual current circuit-breaker 4P 16A 30mA<br>-F1 | 1 | pcs | not priced | - |
| Circuit-breaker 1P C10<br>-Q3 | 1 | pcs | not priced | - |
| ABB T2S160 MA 20<br>-Q4 | 1 | pcs | not priced | - |
| ABB A30<br>-Q5 -Q9 | 2 | pcs | not priced | - |
| ABB TA25DU19<br>-F2 | 1 | pcs | not priced | - |
| ABB T2S160 MA80<br>-Q6 | 1 | pcs | not priced | - |
| ABB A63<br>-Q7 -Q8 | 2 | pcs | not priced | - |
| ABB TA75DU42<br>-F3 | 1 | pcs | not priced | - |
| Bussmann 170M1311<br>-F4 | 1 | pcs | not priced | - |
| ABB ACS880-01-12A6-3<br>-T1 | 1 | pcs | not priced | - |
| Cable 3G1.5 Cu PVC<br>-W1 | 10 | m | not priced | - |
| Cable 4G1.5 Cu PVC<br>-W2 -W4 | 50 | m | not priced | - |
| Cable 7G6 Cu PVC<br>-W3 | 30 | m | not priced | - |

**Incomplete: these lines have no price and are not in the total:**

circuit_breaker:4P:C100 (no price)

circuit_breaker:3P:C16 (no price)

residual_current_device:4P:16A:30mA (no price)

circuit_breaker:1P:C10 (no price)

T2S160 MA 20 (no price)

A30 (no price)

TA25DU19 (no price)

T2S160 MA80 (no price)

A63 (no price)

TA75DU42 (no price)

170M1311 (no price)

ACS880-01-12A6-3 (no price)

cable:3G1.5:Cu:PVC (no price)

cable:4G1.5:Cu:PVC (no price)

cable:7G6:Cu:PVC (no price)

| | |
|---|---|
| Materials | 0.00 |
| Labour and enclosure | 0.00 |
| Markup | 0.00 |
| VAT | 0.00 |
| **Total (JOD)** | **0.00** |

## *Findings (0 blocking, 9 warning, 10 info)*

[WARNING] PP1 / c2: Duty pump (test): Ir 15.2 A, the typical Ir of a 7.5 kW motor in ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3; check it against the nameplate. (motor_current_table; source:

policy; engineering confirmation required)

[WARNING] PP1 / c3: Standby pump (test): Ir 56 A, the typical Ir of a 30 kW motor in ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3; check it against the nameplate. (motor_current_table; source:

policy; engineering confirmation required)

[WARNING] PP1 / c4: Booster (test): Ir 10.38 A from 5.5 kW shaft power at cos phi 0.85, efficiency 0.9; check it against the nameplate. (motor_current_formula; source: policy; engineering confirmation required)

[WARNING] PP1 / c4: Booster (test): the fuses need at least 75 A prospective short-circuit current at the board to operate fast enough. (drive_fuse_fault_level; source: policy; engineering confirmation required)

[WARNING] PP1: Loads without a power factor were taken at cos phi 0.9. (default_power_factor; source: policy; engineering confirmation required)

[WARNING] PP1: No Ze is given for this board, so circuits not under a residual current device are not checked for disconnection on an earth fault; enter the measured or declared Ze. (earth_fault_no_ze; source: policy;

engineering confirmation required)

[WARNING] PP1: Each cable's withstand $k^2S^2$ is in the cable list; compare it with the selected breaker's let-through $I^2t$ at 10 kA from its maker's curve (ABB handbook §2.4). (short_circuit_withstand; source: policy;

engineering confirmation required)

[WARNING] PP1: Circuit rules are this software's defaults, not confirmed by the company's engineers. (rules_unconfirmed; source: policy; engineering confirmation required)

[WARNING] PP1: Enclosure: none named, since 11 devices have no width (rail_widths_mm in the company settings). (enclosure_unknown_widths; source: policy; engineering confirmation required)

[INFO] PP1 / c4: Booster (test): the cable after a drive is protected by the drive's motor overload setting, which is not checked here. (integrity_drive_cable_unchecked; source: policy)

[INFO] PP1: Group 1 breaker raised to 16 A, 1.6 x the 10 A breaker after it, for discrimination. (discrimination_group_raised; source: policy)

[INFO] PP1 / c2: Duty pump (test): dol starter, Type 2 coordination, ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3 Table 3: 400 V 50 kA DOL Normal Type 2. (starter_table; source: policy)

[INFO] PP1 / c3: Standby pump (test): star_delta starter, Type 2 coordination, ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3 Table 5: 400 V 50 kA Y/D Normal Type 2. (starter_table; source: policy)

[INFO] PP1 / c3: Standby pump (test): star and delta contactors need a mechanical interlock; the motor is fed by six conductors (U1 V1 W1, U2 V2 W2). (star_delta_interlock; source: policy)

[INFO] PP1 / c4: Booster (test): ACS880-01-12A6-3 for normal duty at 30 °C, behind 25 A aR fuses (ABB ACS880-01 hardware manual (3AUA0000078093)). Use a screened, symmetrical motor cable, earthed 360° at both ends.

(drive_selected; source: policy)

[INFO] PP1: Breakers are rated for at least 10 kA breaking capacity against 10 kA prospective at the board; each RCCB's conditional short-circuit current with the breaker ahead of it must reach it too.

(breaking_capacity; source: policy)

[INFO] PP1: Leave 1 spare outgoing ways (20 %). (spare_ways; source: policy)

[INFO] PP1: Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip

(about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables. (discrimination; source: policy)

[INFO] PP1: Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation. (cable_conditions; source: policy)